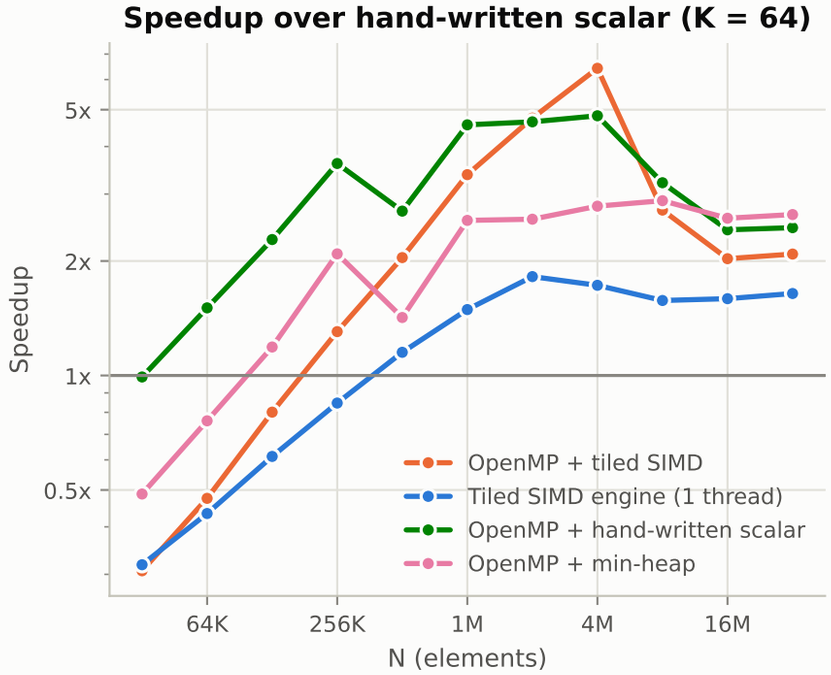
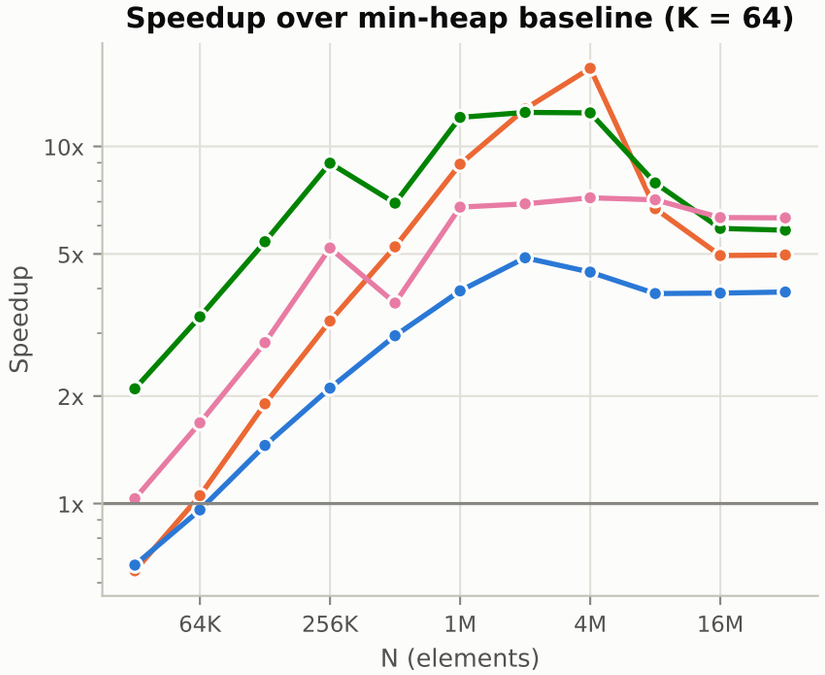
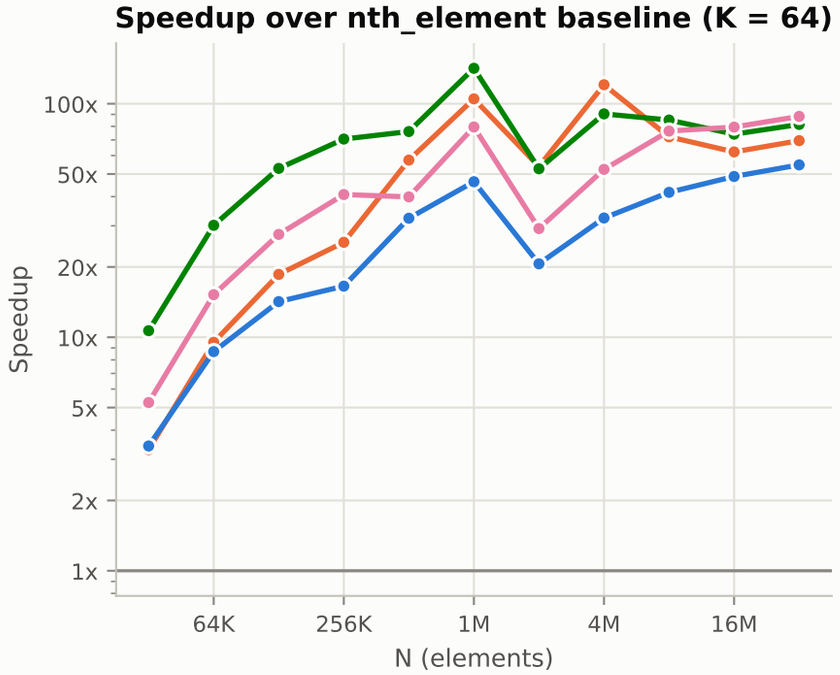

Speedup over hand-written scalar (K = 64)
Speedup
5x
2x
1x
0.5x
OpenMP + tiled SIMD
Tiled SIMD engine (1 thread)
OpenMP + hand-written scalar
OpenMP + min-heap
64K
256K
1M
4M
16M
N (elements)


Speedup over min-heap baseline (K = 64)
Speedup
10x
5x
2x
1x
64K
256K
1M
4M
16M
N (elements)


Speedup over nth_element baseline (K = 64)
Speedup
100x
50x
20x
10x
5x
2x
1x
64K
256K
1M
4M
16M
N (elements)